

Lecture 13: VWAP Strategies

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13.1 Risk-Averse Execution Desk Objective

Intraday market depth follows a pronounced U-shaped volume profile. Executing at a constant rate incurs severe price impact during thin trading periods. To navigate this, institutional desks scale execution with market depth $v_t > 0$ and benchmark performance against the **Volume-Weighted Average Price (VWAP)** [1, 2].

Consider a risk-averse execution desk liquidating initial inventory $Q_0 = \mathfrak{N}$ over $[0, T]$ subject to the terminal constraint $Q_T = 0$. The desk faces two competing costs,

1. **Volume-Scaled Temporary Impact Cost:** $k \frac{\nu_t^2}{v_t}$, where $k > 0$ and $\nu_t = -\dot{Q}_t$ is the execution rate.
2. **Inventory Volatility Risk Penalty:** ϕQ_t^2 , where $\phi \geq 0$ parameterizes risk aversion.

Assuming the unperturbed price is an $(\mathcal{F}_t)$ -martingale independent of volume noise, permanent impact cost is path-independent. Thus, the general objective of the risk-averse execution desk reduces to minimizing total expected cost,

$$\min_{\nu \geq 0} \mathbb{E} \left[\int_0^T \left(k \frac{\nu_t^2}{v_t} + \phi Q_t^2 \right) dt \right] \quad \text{subject to} \quad \int_0^T \nu_t dt = \mathfrak{N} . \quad (13.1)$$

Below, we analyze this objective under three fundamental information regimes.

13.2 Example 1: The Oracle Case ($\mathcal{F}_T$ -Measurable / Anticipating Control)

Suppose the trader possesses perfect foresight of realized total market volume $V_T = \int_0^T v_t dt$ at $t = 0$ (an $\mathcal{F}_T$ -measurable control), and operates as a risk-neutral desk ($\phi = 0$).

Proposition 1 (First-Best VWAP Lower Bound). *Pathwise for every realization $\omega \in \Omega$, the cost-minimizing execution speed is strictly proportional to instantaneous market volume v_t ,*

$$\nu_t^* = \frac{\mathfrak{N}}{V_T} v_t .$$

Proof. Applying the Cauchy-Schwarz inequality for integrals,

$$\mathfrak{N}^2 = \left(\int_0^T \frac{\nu_t}{\sqrt{v_t}} \cdot \sqrt{v_t} dt \right)^2 \leq \left(\int_0^T \frac{\nu_t^2}{v_t} dt \right) \left(\int_0^T v_t dt \right) = \left(\int_0^T \frac{\nu_t^2}{v_t} dt \right) V_T .$$

Multiplying by k yields the absolute pathwise lower bound on execution cost,

$$\int_0^T k \frac{\nu_t^2}{v_t} dt \geq k \frac{\mathfrak{N}^2}{V_T} .$$

Equality holds if and only if $\frac{\nu_t}{\sqrt{v_t}} = c\sqrt{v_t} \implies \nu_t = cv_t$. Integrating over $[0, T]$ yields $c = \frac{\mathfrak{N}}{V_T}$, establishing the exact VWAP profile $\nu_t^* = \frac{\mathfrak{N}}{V_T} v_t$. $\square$

13.3 Example 2: The Static Case ($\mathcal{F}_0$ -Measurable / Open-Loop Control)

In reality, total realized volume V_T is unknown at $t = 0$. Suppose the desk must commit upfront to a deterministic, open-loop execution schedule ν_t using only initial distribution information $\mathcal{F}_0$.

Under a risk-neutral setting ($\phi = 0$), the expectation passes directly inside the integral,

$$\mathbb{E} \left[\int_0^T k \frac{\nu_t^2}{v_t} dt \right] = \int_0^T k \nu_t^2 \mathbb{E} \left[\frac{1}{v_t} \right] dt .$$

Proposition 2 (Harmonic Mean Optimality). *Define the **harmonic mean volume profile** u_t and cumulative volume U_T ,*

$$u_t := \frac{1}{\mathbb{E} \left[\frac{1}{v_t} \right]} , \quad U_T := \int_0^T u_t dt .$$

The optimal deterministic execution schedule is given by:

$$\nu_t^* = \frac{\mathfrak{N}}{U_T} u_t .$$

Proof. Applying Cauchy-Schwarz to $\frac{\nu_t}{\sqrt{u_t}}$ and $\sqrt{u_t}$ over $[0, T]$,

$$\mathfrak{N}^2 = \left(\int_0^T \frac{\nu_t}{\sqrt{u_t}} \cdot \sqrt{u_t} dt \right)^2 \leq \left(\int_0^T \frac{\nu_t^2}{u_t} dt \right) U_T \implies \int_0^T k \frac{\nu_t^2}{u_t} dt \geq k \frac{\mathfrak{N}^2}{U_T} .$$

Equality holds when $\nu_t^* = cu_t$, where integrating over $[0, T]$ yields $c = \frac{\mathfrak{N}}{U_T}$. $\square$

Microstructure Remark (Jensen's Inequality Effect). By Jensen's inequality, $f(x) = \frac{1}{x}$ is strictly convex, so $\mathbb{E}[1/v_t] > 1/\mathbb{E}[v_t] \implies u_t < \mathbb{E}[v_t]$. Static open-loop execution does **not** track the standard arithmetic mean volume $\mathbb{E}[v_t]$. Because temporary impact spikes quadratically during thin trading periods, low-volume periods are penalized more heavily, biasing the open-loop schedule toward periods with lower volume uncertainty.

13.4 Example 3: The Fully-Adapted Case ($\mathcal{F}_t$ -Adapted Stochastic HJB Control)

When the desk continuously observes volume v_t in real time ($\mathcal{F}_t$ -adapted information), the problem becomes a continuous stochastic control problem. Assume v_t follows a general diffusion process $dv_t = \mu_v(t, v_t) dt + \sigma_v(t, v_t) dW_t^v$.

We solve the risk-averse objective (13.1) via **Dynamic Programming**. Define the value function

$$h(t, q, v) = \inf_{\nu \geq 0} \mathbb{E}_t \left[\int_t^T \left(k \frac{\nu_s^2}{v_s} + \phi Q_s^2 \right) ds \right], \quad h(T, q, v) = +\infty \cdot \mathbb{I}_{\{q \neq 0\}}.$$

Hamilton-Jacobi-Bellman (HJB) Equation. By the Dynamic Programming Principle, $h(t, q, v)$ satisfies

$$\partial_t h + \mu_v(t, v) \partial_v h + \frac{1}{2} \sigma_v^2(t, v) \partial_{vv} h + \phi q^2 + \min_{\nu \geq 0} \left\{ k \frac{\nu^2}{v} - \nu \partial_q h \right\} = 0. \quad (13.2)$$

Differentiating inside the minimization operator yields the optimal feedback speed

$$2k \frac{\nu^*}{v} - \partial_q h = 0 \implies \nu^*(t, q, v) = \frac{v}{2k} \partial_q h(t, q, v).$$

Substituting ν^* back into (13.2) gives the non-linear PDE

$$\partial_t h + \mu_v \partial_v h + \frac{1}{2} \sigma_v^2 \partial_{vv} h + \phi q^2 - \frac{v}{4k} (\partial_q h)^2 = 0.$$

Riccati Reduction & The VWAP–TWAP Trade-Off. Testing the quadratic Ansatz $h(t, q, v) = A(t, v)q^2$ yields $\partial_q h = 2A(t, v)q$, and the feedback policy simplifies to

$$\nu_t^* = \frac{v_t}{k} A(t, v_t) Q_t^*.$$

For a deterministic volume profile ($\sigma_v = 0$), $A(t)$ satisfies a non-linear Riccati ODE,

$$\dot{A}(t) = \frac{v_t}{k} A(t)^2 - \phi, \quad \lim_{t \rightarrow T^-} A(t) = +\infty. \quad (13.3)$$

The Riccati equation (13.3) illustrates the explicit structural trade-off faced by the desk:

- **Risk-Neutral Limit** ($\phi = 0$): $\dot{A}(t) = \frac{v_t}{k} A(t)^2 \implies A(t) = \frac{k}{\int_t^T v_s ds}$. The feedback control becomes $\nu_t^* = \left(\frac{v_t}{\int_t^T v_s ds} \right) Q_t^*$, recovering **Pure VWAP tracking** ($\nu_t^* \propto v_t$).
- **High Risk Aversion** ($\phi \gg 0$): Holding inventory incurs severe penalties. The penalty term $-\phi$ increases $A(t)$, forcing the desk to execute faster in early quiet periods and flattening volume participation toward uniform **TWAP execution**.

References

- [1] T. Kato. Vwap execution as an optimal strategy. *JSIAM Letters*, 7:33–36, 2015. arXiv:1408.6118.
- [2] A. Papanicolaou, H. Fu, P. Krishnamurthy, B. Healy, and F. Khorrami. An optimal control strategy for execution of large stock orders using LSTMs. *arXiv preprint arXiv:2301.09705*, 2023.